

SUB CHECK Customer Development

Technology Entrepreneurship • Assignment 3 • Group CS-2406

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Executive summary

SUB-CHECK is a proposed B2B service for independent and AA studios operating competitive shooters. It would analyse aiming behaviour from authoritative game servers, return a risk score with supporting events, and help moderators review suspicious cases alongside existing anti-cheat tools.

Our customer development study provides an early signal of a relevant problem and interest in the concept. In the survey, 17 of 19 participants rated cheating seriousness at 4 or 5, and 16 of 20 rated the proposed server-side system's usefulness at 4 or 5. Respondents nevertheless highlighted detection accuracy, false positives and telemetry privacy. These findings support a controlled pilot, rather than a claim of proven demand or technical effectiveness.

The updated idea retains a studio customer and human enforcement decisions. It adds a self-hosted option for evaluation, explicit telemetry controls, a free pilot and evidence about detection quality. Deployment preferences in the technical section favour self-hosting (6 of 9). Pricing responses favour a free or open-source core with paid cloud services (12 of 20), but this mixed audience does not establish a studio budget or willingness to pay.

Research scope

Evidence	Available material
Online survey	20 participants shown in Google Forms summaries; 27 screenshots, including 2 repeats
Recorded interviews	3 supplied recordings; analysed separately from survey totals
Question-specific bases	19 for seriousness and satisfaction; 9 for technical questions; otherwise 20
Assignment minimum	30 online respondents OR 5 face-to-face recorded video interviews. The current evidence is below both thresholds.

We do not add interview participants to the survey total because overlap is unknown. The results describe this sample and guide the next experiments. They do not estimate the wider gaming market.

Audience and interview preparation

Segmentation and priority

We distinguish the buyer, the operational user and the player affected by enforcement. The scoring below is a team judgement for recruitment planning, on a 1–5 scale. Volume scores are relative estimates, not measured market sizes. Buyer relevance takes precedence when choosing the next interviews.

Segment	Access	Volume	Benefit	Priority
Indie or AA shooter studio leads	3	3	5	1 for purchase validation
Server moderators and tournament administrators	4	3	5	2 for workflow validation
Active competitive players	5	5	4	1 for accessible problem discovery

The reachable survey sample differs from this intended market. Roles are multiple selection: 10 of 20 selected cybersecurity specialist/student, 7 software developer, 7 casual gamer, 3 competitive gamer, 2 game developer, and none esports organiser/admin. Nine reported experience developing, managing, moderating or hosting a multiplayer game/server. Those nine are a technical-experience subgroup, not nine confirmed studio buyers.

Research objectives and question design

Our objectives were to identify customer jobs, pains and desired gains, test reactions to server-side risk scoring, and explore integration and business-model preferences. The questionnaire separates respondent background, cheating experiences, existing solutions, kernel access, the SUB-CHECK concept, risk scoring, technical integration, pricing and final feedback.

Problem questions ask about actual cheating encounters and experience managing games. Concept, trust and pricing questions capture stated preferences after presenting the idea. They are useful for prioritisation but can encourage agreeable answers and do not demonstrate behaviour, purchasing intent or technical results.

Interview guide and recruitment plan

For follow-up interviews, ask everyone about the same core topics before describing SUB-CHECK: their role and recent game; the last suspected cheating incident; what they did; time spent reviewing or appealing; the current tools and their shortcomings; and what outcome would improve their work. Ask technical leads about a recent integration, its effort, data restrictions and actual budget authority. Only then show the concept and ask what evidence would justify a pilot.

Recruit studio leads through existing developer contacts and small multiplayer communities, moderators through server communities, and competitive players through peers. These are proposed channels, not a record of how the existing respondents were recruited. Record permission, role, duration and the source reference. Aim for 15–30 minutes, with neutral probes, no argument and no sales pitch.

Cheating problems and existing alternatives

Observed problem

Question and base	Result	Interpretation
Cheating seriousness, n=19	17 rated 4–5 (89.5%); mean 4.42/5	Strong perceived problem in this sample
Encounter frequency, n=20	4 frequently, 10 sometimes (70% combined)	Past experience supports relevance
Existing solution satisfaction, n=19	6 rated 1–2; 8 rated 3; 5 rated 4–5; mean 2.95/5	Moderate satisfaction, not universal rejection
Biggest current problem, n=20	9 bypasses (45%); 4 false bans (20%); 3 privacy (15%)	Effectiveness and fair enforcement both matter

Wallhacks (15/20) and aimbots (14/20) lead the multiple-selection list of problematic cheat types. Bots received 10 selections, macros/scripts and exploits 9 each, and hardware-assisted cheats 7. SUB-CHECK's aim-analysis scope addresses part of this list. It does not justify a promise to detect wallhacks, bots or every hardware cheat.

Customer jobs and pains

The player's functional job is to participate in fair matches. Cheating interrupts that goal and makes results less trustworthy. The studio and moderator jobs remain hypotheses to validate with buyers: investigate reports efficiently, make defensible enforcement decisions and handle appeals without excessive work. The interview analysis on page 5 adds examples of actual behaviour to the survey summaries.

Survey feedback points to two linked pains: existing tools can be bypassed, while incorrect sanctions can harm legitimate players. This means a useful product must improve the quality of evidence and keep a review process. A high volume of alerts alone would not demonstrate value.

Kernel access and platforms

Eight of 20 would probably not or would not install a kernel-level anti-cheat, six were unsure, and six would or probably would. The largest individual concerns were security vulnerabilities (9/20) and privacy (7/20). These answers indicate friction, without proving that all customers reject kernel tools.

Windows received 17 selections, Linux 5 and SteamOS/Steam Deck 4. Platform selections overlap, so Linux and SteamOS cannot be combined into a unique participant count. Compatibility remains a relevant secondary benefit, while the immediate pilot should focus on reliable analysis in one supported game.

Sources: survey screenshots Q05–Q12 and Q04. Multiple-selection percentages use the number answering that question, not the total number of selections.

Concept feedback and adoption conditions

Useful concept with unresolved accuracy concerns

Sixteen of 20 rated concept usefulness at 4 or 5 (80%), four rated it 3, and none rated it 1 or 2. The mean is 4.20/5. Positive responses to a described concept are an interest signal. They do not prove detection quality or adoption.

Multiple-selection question, n=20	Selections
Main advantages	No kernel software 12 (60%); better privacy 11 (55%); hardware-cheat detection 10 (50%); platform compatibility 7 (35%)
Main concerns	Detection accuracy 11 (55%); false positives 9 (45%); telemetry privacy 9 (45%); server costs 7 (35%)
Response to suspicious behaviour	Manual review 12 (60%); additional verification 12 (60%); temporary restriction 8 (40%); immediate ban 7 (35%)

All 20 said yes or probably yes to trusting an anti-cheat more when a suspicious player receives review before a permanent ban. This question presents review positively, so the result supports testing a review workflow without showing that respondents will accept its delay or cost. Manual review and verification selections overlap and cannot be added.

Technical integration and data control

In the technical-experience subgroup (n=9), seven rated the difficulty of implementing effective anti-cheat at 4 or 5. Six selected a self-hosted solution and three a managed cloud service. REST API, SDK and game-engine plugin each received four selections. These are overlapping preferences, not mutually exclusive product alternatives.

Five of nine would send anonymised telemetry with strict privacy controls, three said yes, and one was unsure. Self-hosting preference and conditional data sharing can coexist. The available aggregates do not reveal which people selected both. Unreal Engine received four preferences, Unity two, and the remaining three selected Godot, a custom engine or a free-text engine answer. This supports an Unreal-first interview plan, not an immediate commitment to a full plugin.

Pricing and open feedback

Twelve of 20 selected a free/open-source core plus paid cloud service, eight a fixed monthly subscription, six per active player, six per analysed session and four tiered subscription. Fifteen said a free tier would increase their likelihood of testing, and five said maybe. These findings motivate a bounded pilot; price levels and paid conversion remain untested.

Q26 reports nine open answers but only eight are visible. Among the visible comments are “live game ban showcase” and “Cheaters, they make you lose all desire to play”. Other substantive comments request transparency, low overhead, privacy protection and independently audited security. We treat these as qualitative themes without a prevalence estimate. Q27 reports eight answers with seven visible, mostly brief non-additions.

Sources: Q13, Q15–Q18 and Q20–Q27. Q14 repeats Q13 and Q19 repeats Q03.

Recorded interview findings

Three recordings add qualitative context to the survey. We use anonymised IDs, paraphrase the clear passages and retain timestamps for traceability. Interviewees describe different game types, so their comments should not be combined into a single target-market vote.

I01 Multiplayer mobile game director

The respondent describes daily work on cheating and server protection (01:14–02:13). The main protection need concerns key game values such as money and ammunition rather than aim input (03:32–04:12). This is a useful negative fit signal: a studio can have a serious cheating problem without needing SUB-CHECK's proposed aim-analysis product.

The respondent finds the general idea interesting but needs details to evaluate it (04:59–06:18). Integration requires development-lead review, control over code and data, and clear responsibility when a partner handles data (06:47–08:30). These are adoption conditions, not a pilot or purchasing commitment. The remarks about legal obligations are the respondent's requirements, not a legal assessment in this report.

I02 Independent game developer with an online leaderboard

The respondent describes a small game with a leaderboard and a server-side maximum-score filter (00:45–01:23). They describe inspecting logs after the system excluded users incorrectly (01:48–02:20). Their desired gain is reliable validation without errors that reject legitimate results. This supports attention to false positives, while leaderboard integrity is outside the initial aim-analysis scope.

The respondent values platform reach (02:33–02:53) and prefers a configurable SDK over automatic integration, citing control over a personal project (04:11–04:31). Interest in delegation is conditional on having a suitable server-based game (03:25–03:43). No budget or agreement to adopt is recorded.

I03 Developer with multiplayer mod and server experience

The respondent describes modding and small-game development, including multiplayer/server work (01:07–02:51), and discusses adding Minecraft server mods for suspicious behaviour (03:08–03:30). Performance and compatibility are important concerns (05:12–07:30). These are reported experiences, not verified industry claims.

Telemetry must not block gameplay or punish network/service failures (12:00–13:17). Collecting additional data can still add overhead, and retrofitting an existing game differs from integrating during development (15:15–17:17; 21:40–23:56). The respondent favours configurable engine integration for a new game, a usable free core with paid extras (32:25–35:19), and an appeal route for false positives (35:41–38:19). No price or commitment is recorded.

Cross-interview interpretation

The interviews support control, integration reliability and fair handling of uncertainty. They also narrow the segment: prioritise a shooter studio with an actual aim-cheat review problem. Mobile economy protection and leaderboard validation belong in a later discovery backlog. No interview demonstrates model accuracy, measured time savings, a price or a paid commitment.

Sources: I01 audio 09:06; I02 audio 04:46; I03 desktop recording 38:58. Timestamped notes and the evidence manifest map findings to the original recordings. We paraphrase clear passages because automatic transcription contains errors.

Hypotheses and changes to the business idea

What the evidence supports

Hypothesis	Evidence and status
Cheating is a meaningful player problem	Supported within this sample: 17/19 seriousness ratings at 4–5 and 14/20 frequent or occasional encounters
Customers value a server-side approach	Early interest: 16/20 usefulness ratings at 4–5. Adoption is untested
Privacy and kernel access matter	Supported as concerns: 12/20 selected no kernel software as an advantage; 9/20 worry about telemetry privacy
Review increases trust	Stated preference supported: 20/20 yes or probably yes; operational cost and delay untested
Studios will buy a cloud-only product	Unvalidated, with a deployment challenge: 6/9 select self-hosting
Behaviour analysis detects hardware cheats accurately	Unvalidated. The survey measures expectations; no labelled technical benchmark exists
Customers will pay at the proposed rate	Unvalidated. No actual budgets, commitments or price-level experiment

Changes relative to Assignment 2

Assignment 2 already narrowed the customer to an indie or AA shooter studio, positioned the tool beside existing protection and required moderator review. We retain that baseline. The current study changes the delivery and validation priorities rather than presenting those earlier choices as new discoveries.

Before	Updated proposal	Reason
Hosted service first	Evaluate self-hosted deployment alongside managed cloud	6/9 technical respondents select self-hosting
General integration support	Minimal API and event schema first; assess Unreal adapter in buyer interviews	API, SDK and plugin tie at 4/9; Unreal is 4/9
Risk score and dashboard	Supporting events, review trail and appeal workflow in the pilot	Accuracy 11/20; false positives 9/20; review preference 20/20
Telemetry ingestion	Minimise fields; pseudonymous session IDs; retention and deletion controls	Telemetry privacy 9/20; conditional sharing 5/9
Pilot and usage tiers	Free bounded evaluation; test fixed and usage pricing with buyers	Free tier yes 15/20; preferences do not prove payment

The recordings also change qualification and architecture. Mobile game values and leaderboard validation are adjacent problems, so we prioritise actual aim-cheat review needs. Before a pilot, confirm that the studio can export the required data: IO3 distinguishes new-game integration from potentially expensive retrofitting. Use asynchronous analysis and never treat missing telemetry alone as proof of cheating.

We do not promise immunity to hardware cheats, an unbreakable cloud model, calibrated cheat probability or full operating-system compatibility for an entire game. A server-side score is a hypothesis to evaluate per game and should remain a prioritisation signal until the evidence supports a stronger interpretation.

Updated business model

Customer and value proposition

SUB-CHECK is a proposed B2B tool for small shooter studios with authoritative servers, a moderation workflow and feasible access to the required telemetry. It would help moderators prioritise suspicious aim events, inspect evidence and record decisions. The studio integrates and pays; players gain fairer review without an additional SUB-CHECK kernel installation. Faster review and fewer incorrect decisions are intended gains to measure.

Business Model Canvas block	Updated working hypothesis
Customer segments	Indie and AA competitive shooter studios. Technical leads or owners as buyers, moderators as users. Budget authority still unverified
Value proposition	Explainable aim-analysis risk scores, human review, controlled telemetry and a choice of deployment
Channels	Developer communities, technical documentation and direct studio outreach. Channel effectiveness untested
Customer relationships	Assisted pilot integration, feedback sessions and technical support
Revenue streams	Bounded free pilot, followed by tested paid cloud usage or predictable monthly plans; optional self-hosted support. No confirmed prices
Key activities	Telemetry integration, model evaluation, moderator workflow development, privacy controls and integration support
Key resources	Game-specific labelled data, integration expertise, a model pipeline and secure processing infrastructure
Key partners	Pilot studios and moderators supplying permissioned data; engine/community contacts for integrations
Cost structure	Engineering, data labelling, inference and storage, security review and customer support. Unit costs remain unmeasured

Commercial implications

The sample can inform product priorities but cannot establish market size or revenue. Only two respondents selected game developer, and none selected esports organiser/admin. Software or cybersecurity experience does not imply a studio purchasing role. The next commercial decision therefore depends on interviews with people who own a game, an integration decision and a budget.

A free/open-source core is a pricing preference to test, not a licence commitment. Before choosing it, compare support burden, security update responsibility, evaluation-to-paid conversion and whether customers need local deployment because of privacy policy, latency or infrastructure control.

Value proposition and validation roadmap

Updated customer profile and value map

Customer profile	Proposed product response
Job: identify suspicious matches and make defensible decisions	Risk-ranked cases with highlighted aim events and a review log
Pain: bypasses reduce confidence in existing protection	Add behavioural evidence to the existing detection stack; measure coverage by cheat type
Pain: false positives and disputes	Human review before permanent bans; preserve supporting events and an appeal trail
Pain: data exposure and integration uncertainty	Minimal documented schema, pseudonymous IDs and explicit storage/deletion policy
Gain: useful detection with manageable overhead	Benchmark detection, time to review, latency, bandwidth and cost per session
Gain: deployment and evaluation flexibility	Bounded free pilot plus self-hosted and managed options to compare

Next experiments and proposed decision rules

These experiments and thresholds are proposed by the team. They are not outcomes of the current study and should be agreed with pilot customers before collecting data.

Experiment	Evidence to collect	Decision rule
Buyer discovery	At least 8 studio leads or moderators; recent incidents, review effort, budget owner, data constraints	Continue if at least 3 studios offer a concrete pilot with available data and an owner
Silent technical pilot	Labelled clean and cheating sessions, separate train/test players, reviewers blind to model score	No automatic bans; compare precision and false alerts at the customer-agreed threshold
Review workflow test	The same cases with and without ranked evidence; paired review time and decision quality	Target at least 20% shorter median review time without worse decisions
Deployment comparison	Integration time, permitted data, bandwidth, latency and operating cost	Choose default deployment from actual pilot constraints, not survey counts alone
Commercial test	A specific paid offer after a successful free pilot and measured unit costs	Treat payment or signed commitment as stronger evidence than a pricing preference

I03 adds a reliability test: simulate slow or unavailable analysis and packet loss. Gameplay must continue, and missing telemetry alone must not increase a player's risk or cause a sanction. Measure the overhead on the authoritative server rather than assuming a remote service is free of performance cost.

Precision, recall and false alerts require trusted labels and game-specific evaluation. Include high-skill legitimate players, latency variation and different input devices. Restrict access to pilot data and log reviewer decisions. A risk score should not become a permanent-ban trigger during this validation phase.

Separately, close the assignment evidence gap by adding the required participants with verifiable records. Do not retrospectively reconstruct individual survey answers from the aggregate charts.

Limitations conclusions and references

Limitations

The available survey contains 20 participants, below the 30 required online respondents. Three recording files also fall below the five face-to-face video-interview alternative; audio files and a desktop recording should not be described as five face-to-face videos. This affects the interview execution criterion even though the available findings can still support analysis and an updated idea.

We have aggregate screenshots rather than the response export. Therefore we cannot verify participant uniqueness, compare roles with preferences, estimate overlap with interviewees or reconstruct individual response rows. Some questions have 19 responses, and the technical branch has nine. Roles, platforms and several preference questions allow multiple selections. Their percentages can exceed 100% when added.

The original questionnaire introduction and branching settings are not fully visible. The nine-person technical section matches the number answering yes to game/server experience, but this alone does not verify routing. Two screenshots repeat other charts. Two open-response screenshots show fewer answer cards than their displayed totals, so those sections are incomplete.

The small reachable sample contains many software and cybersecurity respondents and few confirmed game developers. Recruitment details and survey dates were not documented in the supplied summaries. Concept questions and feature descriptions may bias respondents toward the proposed solution. No market-size estimate, customer payment, retention effect or working detection benchmark follows from these materials.

Interview notes rely on local automatic transcription, which can misrecognise names and specialised terms. Findings use clear, relevant passages and include timestamps, while exact quotations from the audio are avoided. Uncertain details require source-audio review before being used as stronger evidence.

Conclusion

The study supports continuing SUB-CHECK as a focused tool for studios and moderators, with detection quality and fair review as the immediate priorities. We retain server-side aim analysis alongside existing protection, add data control and deployment choice, and use a free bounded pilot to test integration and operational value. Buyer demand and technical performance remain the central questions for the next stage.

Sources

1. Assignment 3 Business model Customer Development instructions. Local course document. Assessment requirements and supplied methodology examples.
2. SUB-CHECK Assignment 2 report. Local project baseline, pages 1–5. Customer, value proposition and original pilot model.
3. Survey/photo_1 ... photo_27_2026-10-08_14-51-25.jpg. Google Forms aggregate summaries. Question IDs follow screenshot numbers. Aggregate transcription and evidence map accompany this report.
4. Interviews folder. Three supplied recordings, referenced by anonymised IDs I01–I03 and source timestamps in the interview notes.
5. ProductStar. How and why a product manager conducts CustDev. Habr, 3 June 2020. <https://habr.com/ru/companies/productstar/articles/505208/>. Used for role segmentation, neutral questions about past behaviour and evidence-linked insights.
6. The second course-supplied methodology example on vc.ru was inaccessible during review. Its URL and access status appear in sources/README.md; no factual claim relies on it.

SUB-CHECK Customer Development

Server-side aim analysis for studio moderators

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Research scope

20 survey participants, with question bases of 20, 19 or 9

3 recordings, analysed separately from the survey

Aggregate charts support counts, but not individual response reconstruction

The sample remains below 30 online respondents or 5 face-to-face video interviews

Customer roles and recruitment priorities

Buyer: indie or AA shooter studio lead with a budget and integration authority

Operational user: moderator reviewing suspicious matches and appeals

Player: benefits from fair matches and defensible enforcement

Next recruitment priority: confirmed studio buyers and moderators

Cheating is a meaningful problem in this sample

89.5%

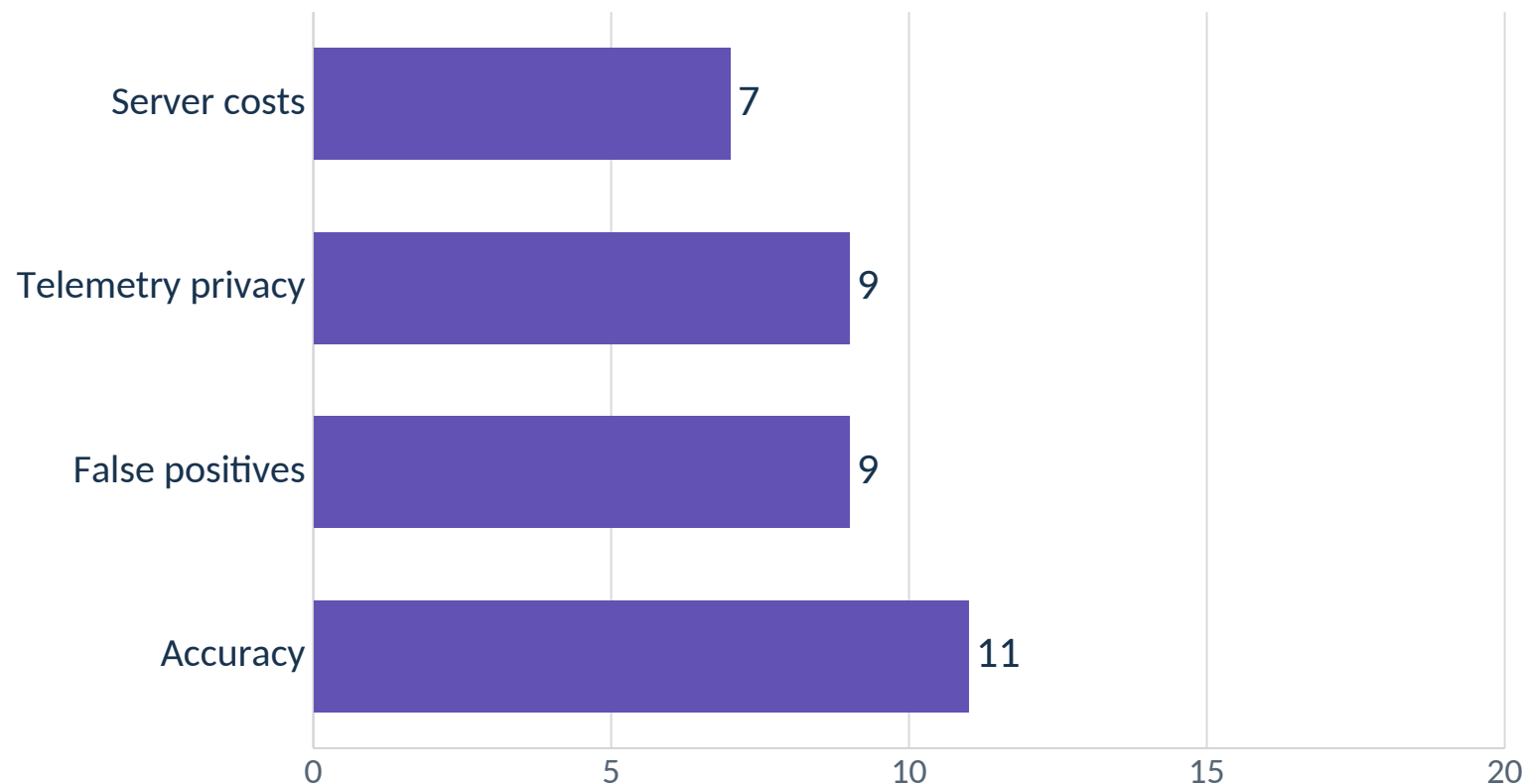
17 of 19 rate cheating seriousness at 4–5

70%

14 of 20 encounter cheating frequently or sometimes

Existing solution satisfaction averages 2.95/5 across 19 answers.

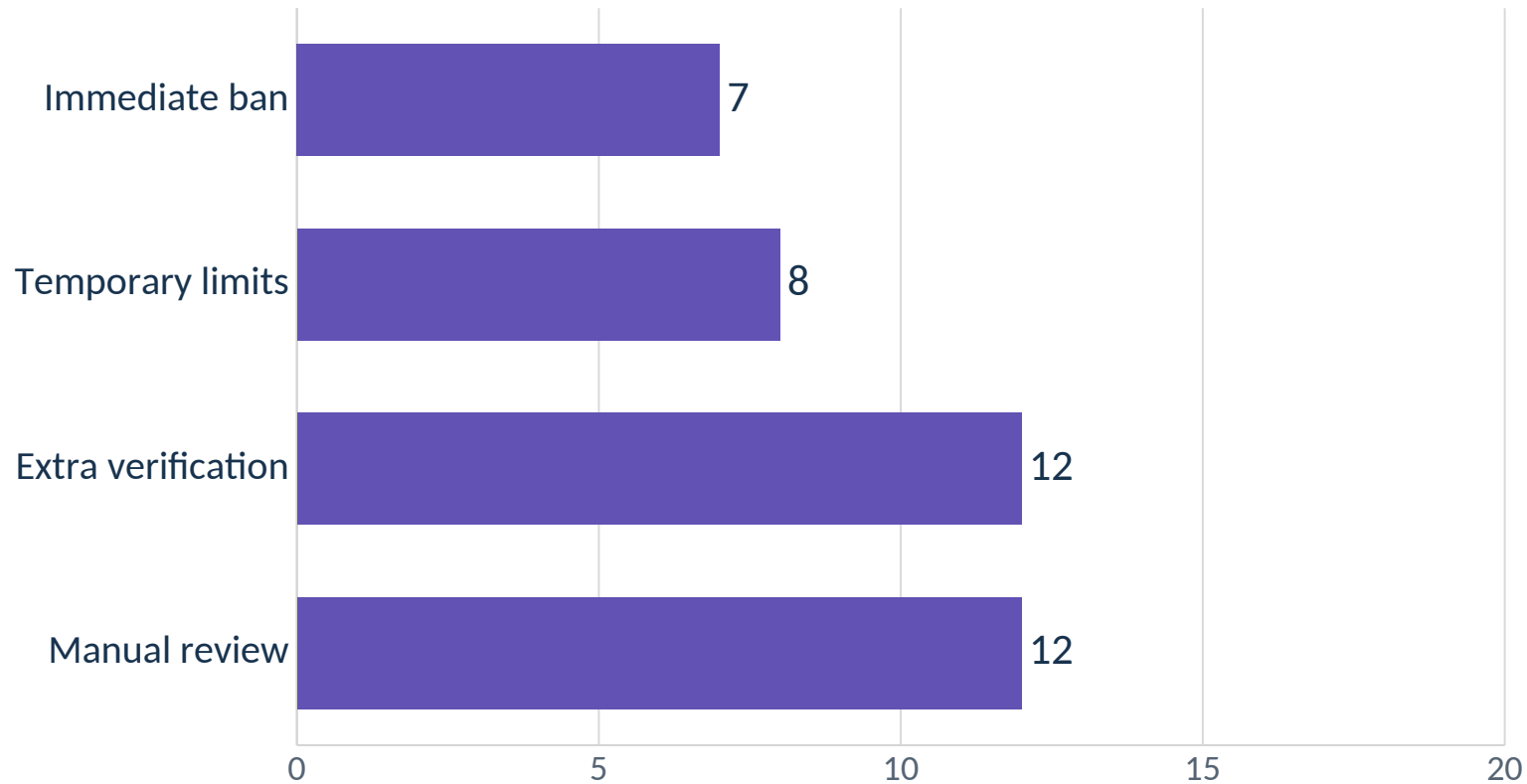
Concept interest and accuracy concerns



16 of 20 rate usefulness at 4–5.

No benchmark or purchase commitment follows from this interest.

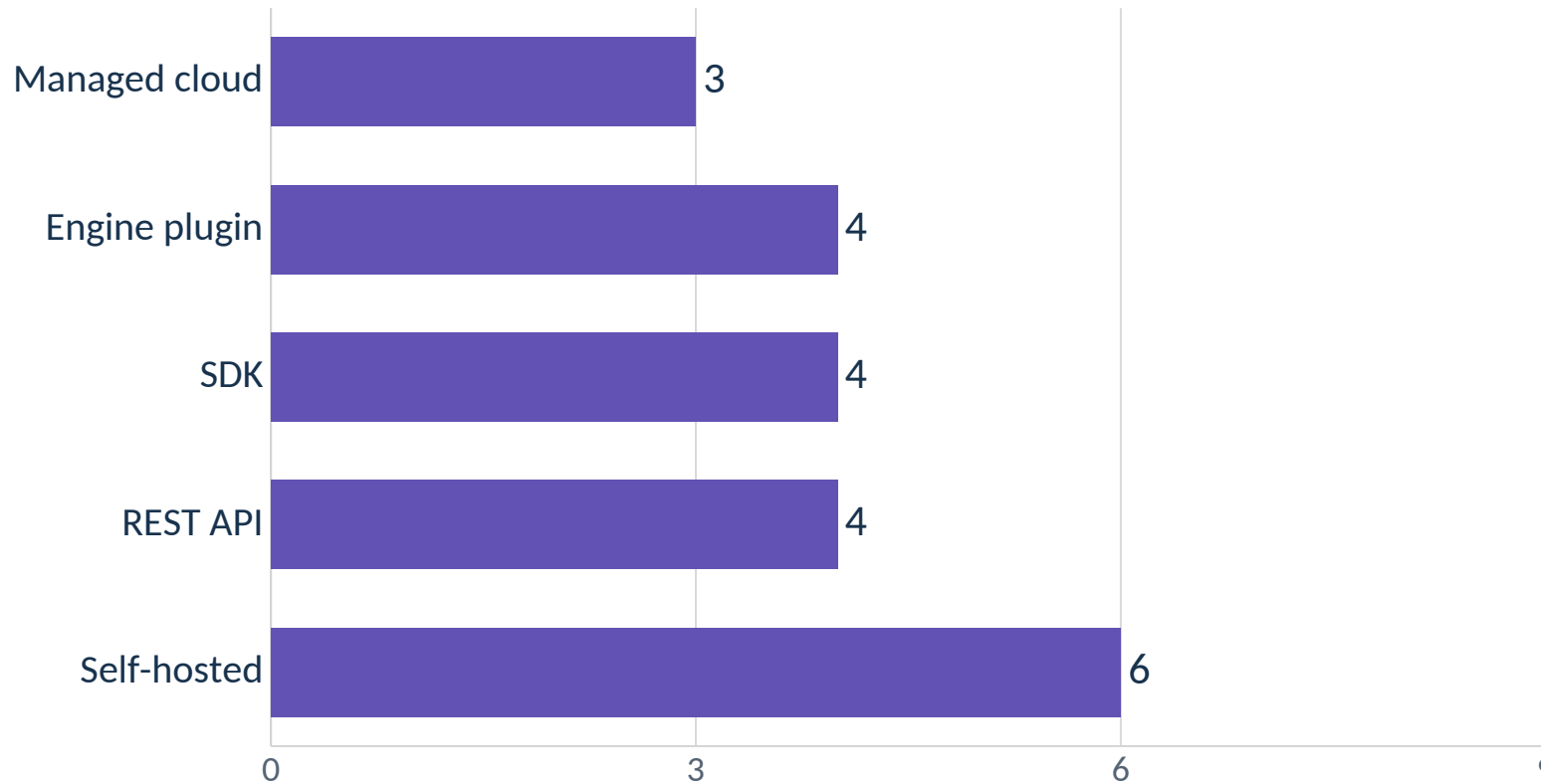
Enforcement preferences



20 of 20 say review before permanent bans would increase trust.

The cost and delay of review remain untested.

Deployment control matters to technical respondents



5 of 9 accept telemetry sharing only with strict privacy controls.

Multiple selections may overlap.

Recorded interviews distinguish different customer problems

I01 mobile director: game values and data control matter more than aim input

I02 indie developer: leaderboard reliability and configurable SDK integration

I03 multiplayer developer: telemetry must not block play or penalise service failures

Scope: prioritise studios with an actual aim-cheat review problem

Business idea updates

Retain supplementary aim analysis and moderator enforcement

Qualify telemetry access and evaluate self-hosted deployment alongside cloud

Add minimal telemetry, explicit retention and a review trail

Use a bounded free pilot and test paid offers with studio buyers

Updated business model

Customer: studio lead buys; moderators use the review workflow

Value: prioritised aim events with evidence and human decisions

Delivery: documented API plus deployment choice and assisted integration

Revenue: test monthly or usage plans after a free pilot; measure operating costs

Pilot validation

Interview at least 8 studio leads or moderators about recent incidents and budgets

Seek 3 concrete studio pilots with a data owner and permitted telemetry

Measure detection quality on held-out players and review every enforcement case

Target 20% shorter median review time without worse decisions

Conclusion and remaining evidence gaps

The evidence supports focused evaluation of SUB-CHECK

Accuracy, false alerts and telemetry control are the immediate priorities

Buyer demand, paid conversion and technical performance remain unvalidated

Additional participant evidence is still needed to meet the assignment minimum